

POLARONS

Emin provides experimental and theoretical graduate students and researchers with a distinctive introduction to the principles governing polaron science. The fundamental physics is emphasized and mathematical formalism is avoided. The book gives a clear guide to how different types of polaron form and the measurements used to identify them. Analyses of four diverse physical problems illustrate polaron effects producing dramatic physical phenomena.

The first part of the book describes the principles governing polaron and bipolaron formation in different classes of materials. The second part emphasizes the distinguishing electronic-transport and optical phenomena through which polarons manifest themselves. The book concludes by extending polaron concepts to address critical aspects of four multifaceted electronic and atomic problems: Large-bipolaron superconductivity, electronic switching of small-polaron semiconductors, electronically stimulated atomic desorption, and diffusion of light interstitial atoms.

DAVID EMIN retired from Sandia National Laboratories and is currently Adjunct Professor in the Department of Physics and Astronomy, University of New Mexico. He is best known for his work on polaron formation and motion and is often cited for his seminal contributions to the theories of self-trapping and hopping conduction's phonon-assisted transition rates, Hall effect and Seebeck effect.

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CAMBRIDGE UNIVERSITY PRESS
Cambridge, New York, Melbourne, Madrid, Cape Town,
Singapore, São Paulo, Delhi, Mexico City

Cambridge University Press
The Edinburgh Building, Cambridge CB2 8RU, UK

Published in the United States of America by Cambridge University Press, New York

www.cambridge.org
Information on this title: www.cambridge.org/9780521519069

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First published 2013

Printed and bound in the United Kingdom by the MPG Books Group

A catalog record for this publication is available from the British Library

Library of Congress Cataloging in Publication data

Emin, D. (David)

Polarons / David Emin.

pages cm

ISBN 978-0-521-51906-9 (hardback)

1. Polarons. 2. Electrons. I. Title.

QC176.8.P62E45 2013

539.7'21—dc23

2012022011

ISBN 978-0-521-51906-9 Hardback

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